Customer Churn Analysis

"Telecom companies lose 1 in 4 customers — this project identifies WHY and HOW to reduce churn

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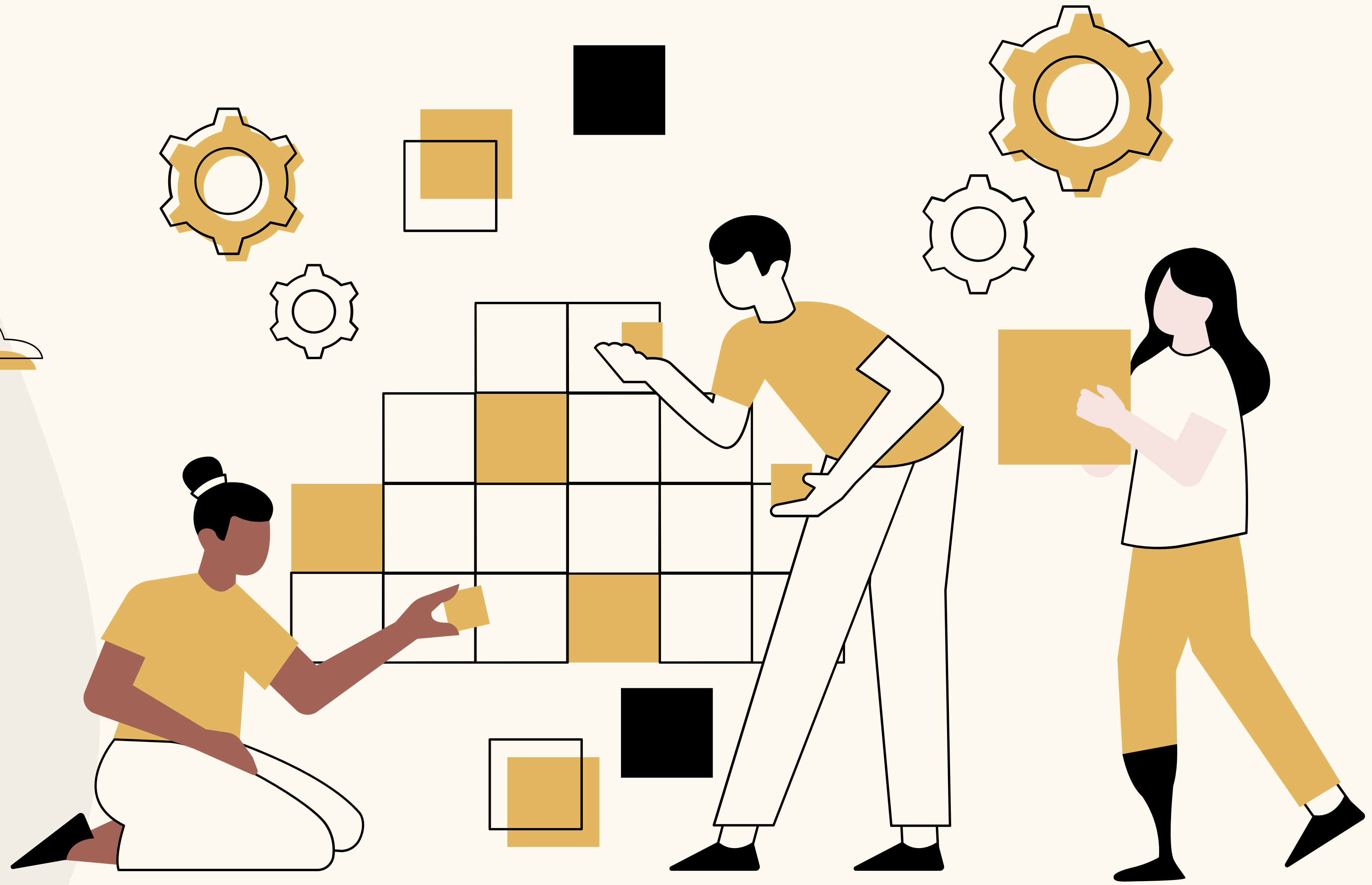

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- Tools Used: MySQL
- Dataset: Telco Customer Churn(Kaggle)

Project Overview

This project analyzes telecom customer data using SQL to identify key factors influencing customer churn and provide actionable business insights.

7,032 Total Customers

Dataset contains 7,032 customers with 21 attributes

26.58% Churn Rate

Approximately 1 in 4 customers discontinued the service, indicating a high churn risk

4 Database Tables

Data normalized into 4 relational tables: Customers, Services, Billing, and Churn Status

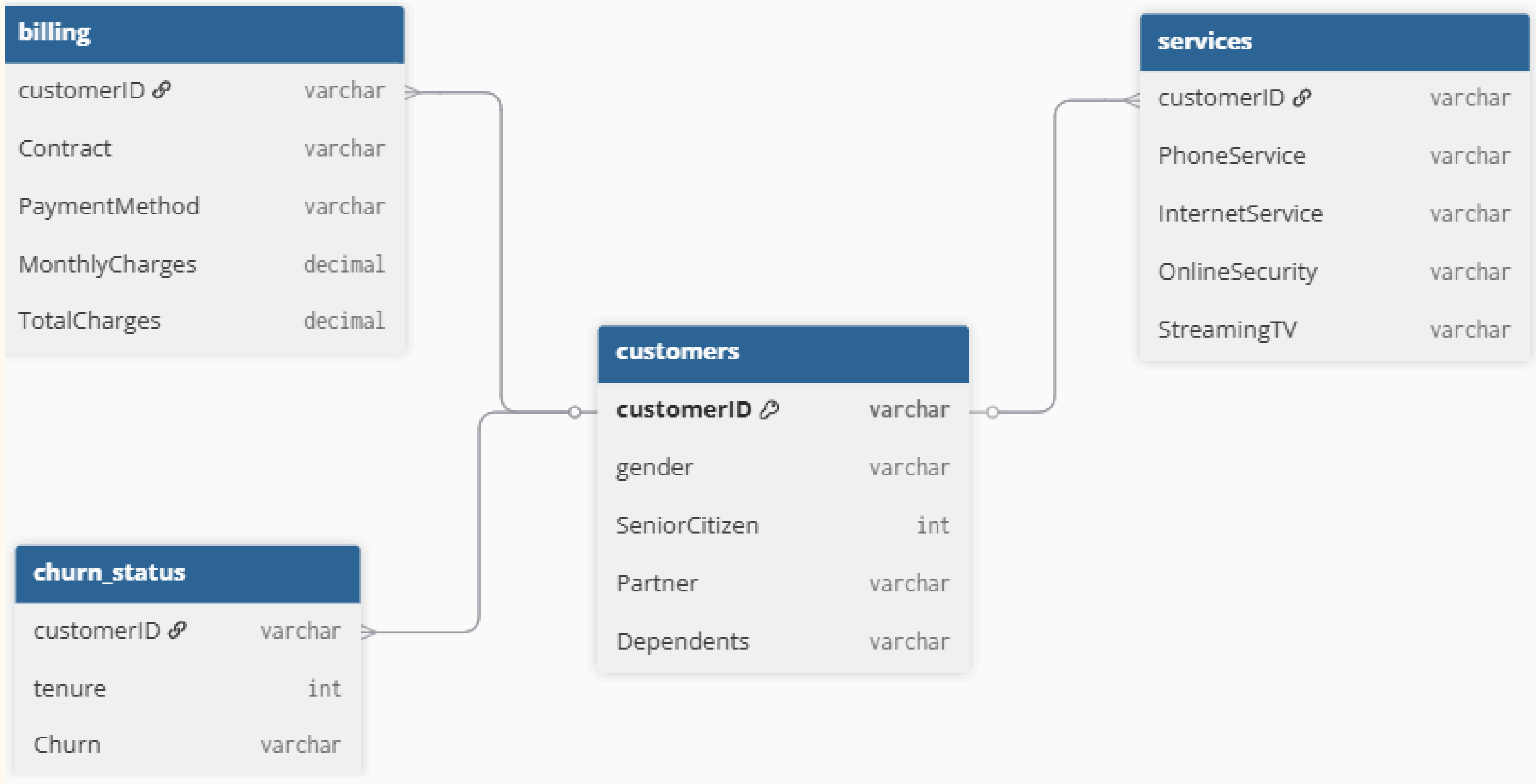
12 SQL Queries

Includes queries ranging from basic exploration to advanced SQL techniques such as CTEs and window functions

Data Normalization & Structure

- Designed and created relational database **telecom_customer**
- Transformed raw dataset into a **normalized structure (3NF)**
- Split single table into **4 logically structured tables**
- Established relationships using customerID **(Primary Key)**
- Ensured data **consistency** and reduced **redundancy**

Normalized Database Schema



Why Normalization?

To eliminate data redundancy, improve data integrity and improve query efficiency

Data Cleaning

Check 1- NULL Values

- Checked all 4 tables for missing values using SUM+CASE WHEN
- Tables checked: customers, services,billing, churn_status

```
select SUM(CASE when customerID is null THEN 1 else 0 END) as null_CustomerID,
SUM(CASE when gender is null THEN 1 else 0 END) as null_gender,
SUM(CASE when SeniorCitizen is null THEN 1 else 0 END) as null_SeniorCitizen,
SUM(CASE when Partner is null THEN 1 else 0 END) as null_Partner,
SUM(CASE when Dependents is null THEN 1 else 0 END) as null_Dependents
from customers;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

null_CustomerID	null_gender	null_SeniorCitizen	null_Partner	null_Dependents
0	0	0	0	0

Check 2 - Duplicates

- Checked for duplicate customer IDs using GROUP BY + HAVING
- Each customer appears exactly once

```
SELECT
    COUNT(customerID) AS Duplicatecustomer_ID
FROM
    customer_churn
GROUP BY customerID
HAVING Duplicatecustomer_ID > 1;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Duplicatecustomer_ID

CLEANING RESULTS

No missing values found across all tables
No duplicate customer IDs detected
Final dataset contains 7,032 clean and reliable records

Dataset is clean, consistent, and ready for analysis.

DATA EXPLORATION & ANALYSIS

“Key Insights Derived from 12 SQL queries”

Key Business Insights

26.58%
OVERALL CHURN
1 in 4 customers left

42.71%
MONTH-TO-MONTH
Highest churn contract

41.89%
Fiber OPTIC
Highest churn service

45.29%
ELECTRONIC CHECK
Highest churn payment

$139K
REVENUE LOST
Total monthly loss

Senior Citizens at High Risk
Senior citizens churn **41.68** almost double non- seniors at 23.65%.Targeted retention needed!

New customers Most Vulnerable
Customers in first 12 months churn at **47.68%**. Early engagement programs are critical!

High Paying Customers Leave
Churned customers pay an avg **$74.44** vs month **$61.31** for loyal customers. Value perception is key!

1.) Overall & Gender-wise Churn Analysis

Overall Churn

```sql
SELECT
    COUNT(customerID) AS total_customers,
    SUM(CASE
        WHEN Churn = 'Yes' THEN 1
        ELSE 0
    END) AS churned_customers,
    ROUND(SUM(CASE
                WHEN Churn = 'Yes' THEN 1
                ELSE 0
            END) * 100 / COUNT(customerID),
            2) AS churn_rate_percent
FROM
    churn_status;
```

Result Grid | Filter Rows: | Export:

total_customers	churned_customers	churn_rate_percent
7032	1869	26.58

Gender-Wise churn

```sql
    c.gender,
    COUNT(*) AS total_customers,
    SUM(CASE
        WHEN churn = 'Yes' THEN 1
        ELSE 0
    END) AS churned_customers,
    ROUND(SUM(CASE
                WHEN churn = 'Yes' THEN 1
                ELSE 0
            END) * 100 / COUNT(*),
            2) AS gender_churn_rate_percent
FROM
    churn_status AS ch
        JOIN
    customers AS c ON c.customerID = ch.customerID
```

gender	total_customers	churned_customers	gender_churn_rate_percent
Female	3483	939	26.96
Male	3549	930	26.20

Key Insights-

- Overall churn rate is **26.58%**, indicating significant customer attrition
- Churn rate is almost identical across genders **(~26%)**
- Gender is not a strong factor influencing churn
- Retention strategies should focus on other variables (contract, tenure, services)

2.) Churn Analysis by Senior Citizens & Dependents

Churn by Senior Citizens

```
SELECT
    c.SeniorCitizen,
    COUNT(*) AS total_customers,
    SUM(CASE
        WHEN churn = 'Yes' THEN 1
        ELSE 0
    END) AS churned_customers,
    ROUND(SUM(CASE
                WHEN churn = 'Yes' THEN 1
                ELSE 0
            END) * 100 / COUNT(*),
            2) AS senior_citizen__churn_rate_percent
FROM
    churn_status AS ch
        JOIN
    customers AS c ON c.customerID = ch.customerID
GROUP BY c.SeniorCitizen;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

SeniorCitizen	total_customers	churned_customers	senior_citizen__churn_rate_percent
0	5890	1393	23.65
1	1142	476	41.68

Churn by Dependents

```
SELECT
    c.dependents,
    COUNT(*) AS total_customers,
    SUM(CASE
        WHEN churn = 'Yes' THEN 1
        ELSE 0
    END) AS churned_customers,
    ROUND(SUM(CASE
                WHEN churn = 'Yes' THEN 1
                ELSE 0
            END) * 100 / COUNT(*),
            2) AS dependents__churn_rate_percent
FROM
    churn_status AS ch
        JOIN
    customers AS c ON c.customerID = ch.customerID
GROUP BY c.dependents;
```

dependents	total_customers	churned_customers	dependents__churn_rate_percent
Yes	2099	326	15.53
No	4933	1543	31.28

- **Key Insights-**
- Senior citizens churn more **(~41.68%)**
- No dependents → higher churn **(~31%)**
- Dependents → better retention
- High-risk group: seniors & independent customers

3.) Churn Analysis by Internet Service & Contract Type

Churn by internet service

```sql
SELECT
    s.InternetService,
    COUNT(*) AS total_customers,
    SUM(CASE
    ROUND(SUM(CASE
                WHEN churn = 'Yes' THEN 1
                ELSE 0
            END) * 100 / COUNT(*),
            2) AS InternetService__churn_rate_percent
FROM
    churn_status AS ch
        JOIN
    services AS s ON s.customerID = ch.customerID
GROUP BY s.InternetService
ORDER BY InternetService__churn_rate_percent DESC;
```

InternetService	total_customers	churned_customers	InternetService__churn_rate_percent
Fiber optic	3096	1297	41.89
DSL	2416	459	19.00
No	1520	113	7.43

Churn by Contract Type

```sql
SELECT
    b.contract,
    COUNT(*) AS total_customers,
    SUM(CASE
        WHEN churn = 'Yes' THEN 1
        ELSE 0
    END) AS churned_customers,
    ROUND(SUM(CASE
                WHEN churn = 'Yes' THEN 1
                ELSE 0
            END) * 100 / COUNT(*),
            2) AS contract__churn_rate_percent
FROM
    churn_status AS ch
        JOIN
    billing AS b ON ch.customerID = b.customerID
GROUP BY b.contract
ORDER BY contract__churn_rate_percent DESC;
```

contract	total_customers	churned_customers	contract__churn_rate_perc
Month-to-month	3875	1655	42.71
One year	1472	166	11.28
Two year	1685	48	2.85

Key Insights-

- Fiber optic users show the highest churn among internet services
- Month-to-month contracts have significantly higher churn rates
- Customers with long-term contracts (1–2 years) show much lower churn
- High churn is driven by flexible plans combined with premium services
- Contract duration plays a stronger role in retention than service type

4.) Churn Analysis by Payment Method & Monthly charges

Churn by Payment method

```
SELECT
    c.dependents,
    COUNT(*) AS total_customers,
    SUM(CASE
        WHEN churn = 'Yes' THEN 1
        ELSE 0
    END) AS churned_customers,
    ROUND(SUM(CASE
            WHEN churn = 'Yes' THEN 1
            ELSE 0
        END) * 100 / COUNT(*),
        2) AS dependents__churn_rate_percent
FROM
    churn_status AS ch
        JOIN
    customers AS c ON c.customerID = ch.customerID
GROUP BY c.dependents;
```

paymentmethod	total_customers	churned_customers	Paymentmethod__churn_rate_percent
Electronic check	2365	1071	45.29
Mailed check	1604	308	19.20
Bank transfer (automatic)	1542	258	16.73
Credit card (automatic)	1521	232	15.25

Churn by Monthly chargese

```
SELECT
    ch.churn,
    ROUND(AVG(b.monthlycharges), 2) AS average_monthly_charges,
    MAX(b.monthlycharges) AS maximum_billing_charges,
    MIN(b.monthlycharges) AS minimum_monthly_charges
FROM
    churn_status AS ch
        JOIN
    billing AS b ON b.customerID = ch.customerID
GROUP BY ch.churn;
```

churn	average_monthly_charges	maximum_billing_charges	minimum_monthly
No	61.31	118.75	18.25
Yes	74.44	118.35	18.85

Key Insights-

- Electronic check has highest churn (~**45%)**
- Auto-payment methods have lower churn
- High monthly charges linked to higher churn

6.) Churn Analysis by Tenure Groups (Using CTE)

Customer Segmentation using CTE

```
with Tenure_Group as(Select churn,customerID,(case when Tenure between 0 and 12 then "New_Customer"
when Tenure between 13 and 24 then 'Mid_Term_Customer'
 when Tenure between 25 and 48 then 'long_term_Customer'
  else 'Loyal_Customer' End)  as Tenure_group  from churn_status)
```

Tenure_group	Total_customer_id	churned_customer	tenure_group_churn
New_Customer	2175	1037	47.68
Mid_Term_Customer	1024	294	28.71
Loyal_Customer	2239	213	9.51
long_term_Customer	1594	325	20.39

Churn Rate by Tenure Group

```
 SELECT
    Tenure_Group,
    COUNT(customerID) AS Total_customer_id,
    SUM(CASE
        WHEN churn = 'Yes' THEN 1
        ELSE 0
    END) AS churned_customer,
    ROUND(SUM(CASE
                WHEN churn = 'Yes' THEN 1
                ELSE 0
            END) * 100 / COUNT(customerID),
            2) AS tenure_group_churn
FROM
    Tenure_Group
GROUP BY Tenure_group;
```

Key Insights-

- New customers have the highest churn **(~47%)**
- Churn decreases as tenure increases
- Mid-term customers show moderate churn **(~28%)**
- Loyal customers have the lowest churn **(~9%)**
- Long-term customers still show notable churn (**~20%)**
- Strong onboarding and early engagement are critical for retention

5) Revenue Loss & Tenure Analysis

Cumulative Revenue Loss by Tenure

```
select ch.tenure,Round(sum(b.monthlycharges),2) as revenue_lost,
Round(sum(sum(b.monthlycharges)) over (order by ch.tenure),2) as cumulative_revenue_lost from churn_status as ch
 join billing as b
 on b.customerID=ch.customerID
 where ch.churn='Yes'
 group by ch.tenure
 order by ch.tenure;
```

tenure	revenue_lost	cumulative_revenue_lost
1	22115.00	22115.00
2	8108.70	30223.70
3	6205.00	36428.70
4	5862.75	42291.45
5	4561.25	46852.70
6	3043.40	49896.10
7	3833.95	53730.05
8	3182.65	56912.70

Tenure Analysis

```
SELECT
    AVG(Tenure) AS average_tenure,
    MAX(Tenure) AS maximum_tenure,
    MIN(Tenure) AS minimum_tenure
FROM
    churn_status;
```

Result Grid | Filter Rows: | Exp

average_tenure	maximum_tenure	minimum_tenure
32.4218	72	1

Key Insights-

- Revenue loss grows cumulatively with churn over time
- Early-stage churn drives the highest revenue impact
- Low-tenure customers are the primary source of revenue leakage
- Avg tenure (**~32 months)** reflects moderate customer retention
- Strengthening early engagement can significantly reduce losses

7.) High-Value Customers Churn Analysis

Top 10 Churned Customers by Monthly Charges

```
select b.customerID,b.monthlycharges,Rank() over(order by monthlycharges desc) as churn_rank
 from billing as b
 join churn_status as ch
 on b.customerID=ch.customerID
 where churn='Yes'
 group by b.customerID
 order by b.monthlycharges desc
 limit 10;
```

customerID	monthlycharges	churn_rank
8199-ZLLSA	118.35	1
2889-FPWRM	117.80	2
2302-ANTDP	117.45	3
9053-JZFKV	116.20	4
1444-VVSGW	115.65	5
0201-OAMXR	115.55	6
4361-BKAXE	114.50	7
1555-DJEQW	114.20	8
9158-VCTQB	113.60	9
7279-BUYWN	113.20	10

Key Insights-

- Churned customers with higher monthly charges lead to major revenue loss
- A small group of high-value customers contributes disproportionately to churn impact
- Business should prioritize retention of high-paying customers

Recommendations

1) Push customers toward longer contracts

What we found: Nearly 4,000 customers are on monthly plans — and almost half of them leave. Customers on 2-year plans? Only 2.85% churn.

Give monthly customers a reason to switch — a small discount or free month if they commit to a yearly plan. This one change could save the most customers.

Est. recovery: $28K+/month

2) Give senior customers extra attention

What we found: Senior citizens leave at 41.68% — almost twice the rate of younger customers. They are clearly not getting what they need..

Create a simpler plan for seniors — easier billing, a helpline they can actually call, and set them up on auto-pay so they don't have to worry each month

1,142 at-risk customers.

3) Stop losing new customers in month 1

What we found: 1 in 2 new customers leaves within the first year. Month 1 alone costs the company $22,115 in lost revenue every month.

Call new customers within the first week. Check in again at day 30. Most people leave early because they feel ignored — a little contact goes a long way.

Biggest revenue impact

4) Fix the fiber optic experience

What we found: Fiber optic users churn at 41.89% — the highest of any internet type. These are customers paying more, yet leaving the most.

Send these customers a short satisfaction survey. Find out what is actually bothering them — speed, reliability, price — and fix those specific problems before they walk out

High-value segment at risk

5) Move customers off manual payments

What we found: Customers paying by electronic check churn at 45.29%. Customers on auto-pay (card or bank transfer) churn at just 15–17%.

Run a simple campaign — offer one month free or a small bill credit to anyone who switches to auto-pay. It removes the monthly hassle and keeps customers longer

Quick win — easy to execute.

Project Summary

- Analyzed telecom churn data of **7,032 customers** using SQL
- Designed a **normalized database (3NF)** with **4** relational tables
- Identified key churn drivers across customer demographics, service usage, and billing behavior
- Found **high-risk segments** such as month-to-month users, senior citizens, and new customers
- Provided data-driven recommendations to improve customer retention and reduce revenue loss

Thank You

[Explore Full Project on Github]